

Building an Automated Healthcare Recommendation System with RAG and Vietnamese Large Language Models

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Abstract

In the era of digitalization, the convergence of healthcare and information technology is driving transformative changes in how people access health services. The rapid advancement of Artificial Intelligence (AI), particularly Large Language Models (LLMs), has opened new directions for automating medical consultation and medical information classification. However, alongside this considerable potential come substantial technical challenges related to the reliability and accuracy of generated information. This project aims to address a core problem: the lack of an intelligent Vietnamese-language automated healthcare advisory system with the capability to mitigate the risks of misinformation.

1. Introduction

1.1. Research Background

Rising demand for online health information retrieval

In recent years, searching for health-related information on the Internet has become a widespread global practice. Sociological surveys on user behavior indicate that a large proportion of people tend to consult “Doctor Google” or online forums before deciding to visit a healthcare facility. This shift reflects an urgent need for timely, low-cost, and convenient access to medical information [1]. In Vietnam, amid severe overcrowding at tertiary hospitals, this demand has become more pressing than ever. People need an initial triage-oriented tool to help determine the appropriate medical specialty, thereby saving time and reducing the burden on the public healthcare system.

Limitations of current LLMs: the “hallucination” problem

In computer science, LLM hallucination is defined as the phenomenon in which a model generates content that appears logical and grammatically fluent but is entirely factually incorrect or unfaithful to the underlying source data [2]. This stems from the fundamental nature of LLMs: they are probabilistic models that predict the next token based on preceding context, without genuinely “understanding” knowledge or verifying factuality. In medical settings, incorrect advice (e.g., suggesting an improper drug dosage or fabricating a non-existent treatment) can lead to serious consequences for users’ health [3].

Moreover, most widely used LLMs are trained predominantly on English data. When operating in Vietnamese—especially with specialized medical terminology—models often struggle to capture cultural context and language-specific characteristics, resulting in inaccurate recommendations or unnatural phrasing [4]. Therefore, relying solely on the parametric knowledge embedded in pre-trained models is not sufficiently safe for healthcare applications. This is the primary motivation for our project: to explore a technical solution capable of grounding AI-generated answers in reputable medical data sources.

1.2. Objectives

Based on the challenges discussed above, this project sets a core objective: to build and experimentally deploy an automated Healthcare Recommendation System that integrates Retrieval-Augmented Generation (RAG). The project objectives are specified as follows:

Symptom understanding from natural language

The system is designed to take users’ symptom descriptions as free-form text input. Instead of requiring users to select rigid keywords, the system allows them to describe their health conditions in everyday language. Its task is to transform these unstructured descriptions into vector representations (embeddings) that can be effectively processed by computational models.

Specialty triage and practical recommendations

Unlike conventional chatbots that primarily engage in casual conversation, the system output targets two concrete values:

- **Specialty recommendation:** Based on the reported symptoms, the system classifies and recommends an appropriate medical specialty (e.g., Pediatrics, Dermatology, Respiratory Medicine) to help users choose a suitable care pathway.
- **Reference health advice:** The system synthesizes information from the medical knowledge base to provide guidance on lifestyle practices, basic first-aid handling, and necessary warnings.

Quality control and faithfulness

A critical objective of this project is to minimize hallucination as much as possible. We do not focus solely on generating “fluent” answers; more importantly, we prioritize answers that are *correct* and grounded in verified data. The system must be able to accurately retrieve relevant passages from the medical repository (**Retrieval**) and use them as context for response generation (**Generation**), rather than allowing the model to “invent” unsupported content.

Finally, the project clearly defines the system’s scope as an informational support tool intended for recommendations and reference. It does **not** provide definitive medical diagnoses and does **not** replace the role of licensed clinicians. All recommendations are accompanied by a disclaimer advising users to consult healthcare professionals [5].

1.3. Scope

To ensure feasibility and sufficient technical depth, this project limits its scope to the following aspects:

Vietnamese medical data

The project focuses on processing and leveraging medical resources written entirely in Vietnamese. The data consist of two main components:

- **Medical article repository:** Collected from reputable Vietnamese healthcare portals (e.g., Vinmec, Long Châu, Tam Anh Hospital, etc.). Using domestic sources helps

ensure that the provided recommendations align with Vietnam’s epidemiological patterns, climate conditions, and healthcare infrastructure.

- **Question–Answer (QA) dataset:** Sample question–answer pairs between patients and clinicians, enabling the model to learn the advisory writing style and practical problem-solving patterns.

Data preprocessing (cleaning and anonymizing personal information) is also within the project scope to ensure compliance with data ethics requirements [6].

Vietnamese-capable language models

From a technological perspective, the project does not train a new LLM from scratch due to computational constraints. Instead, it focuses on applying and adapting state-of-the-art open-source models that provide strong Vietnamese support.

- For **information retrieval (Retriever)**, the project employs BERT-family models trained on large-scale Vietnamese corpora, notably **PhoBERT**, which serves as a foundational model for capturing Vietnamese semantics [7].
- For **text generation (Generator)**, the project explores and compares the performance of newer LLMs such as **PhoGPT** and **Vistral**. These models are selected based on their instruction-following ability and their capacity to generate fluent, coherent Vietnamese text.

The scope also includes designing the RAG architecture, optimizing the vector database for efficient search, and developing a basic interactive interface for experimental evaluation. Topics such as medical imaging diagnosis (e.g., X-ray, MRI) or genomic data are beyond the scope of this project.

2. Database

The quality of an artificial intelligence system—especially in healthcare—depends fundamentally on the quality of its input data. For a Retrieval-Augmented Generation (RAG) architecture, the database not only serves as training material but also functions as a reference knowledge base that supports real-time information retrieval. Within the scope of this project, we established a rigorous data

collection and processing pipeline to ensure accuracy, broad coverage across medical specialties, and compliance with data ethics standards.

2.1. Data Collection

To address the general medical consultation task, we adopted a data strategy that combines domain-specific knowledge (from medical articles) with real-world conversational context (from Question–Answer pairs). Overall, the project dataset consists of two main components:

1. Medical Article Dataset

We collected and consolidated **77,917** Vietnamese medical articles from the online portals of leading healthcare institutions in Vietnam, as summarized in **Figure 2.1** below.

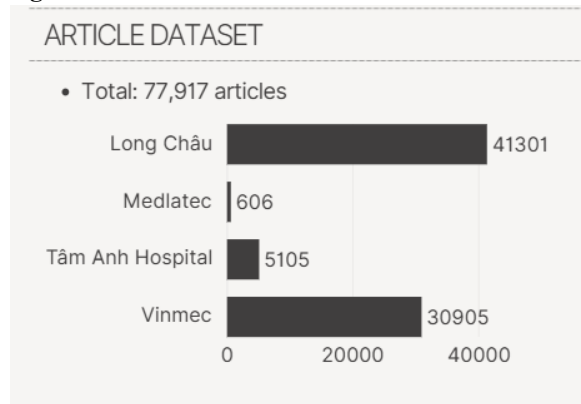


Figure 2.1. Statistics of the medical article dataset.

Source selection was based on criteria of **professional authority** and **information freshness**. Specifically, the articles were collected from the following sources:

- **Long Châu Pharmacy (41,301 articles):** This is the largest source in the corpus, with strong coverage of pharmacological information, medication usage guidelines, dietary supplements, and common conditions. Data from this source plays an important role in enabling the system to answer questions related to drug indications and dosage.
- **Vinmec Healthcare System (30,905 articles) and Tâm Anh General Hospital (5,105 articles):** These are two high-quality clinical sources, authored and reviewed by specialist physicians. Articles from these sources provide in-depth knowledge on diseases, diagnostic procedures, surgery, and

advanced treatment methods. They form the system’s core knowledge backbone when handling complex symptom-related queries.

- **Medlatec (606 articles):** Supplements the corpus with information on laboratory testing and paraclinical diagnostics.

Each collected article was structured into fields including **Title**, **Introduction**, **Headings**, and **Detailed Content**. This structure allows the Retriever to pinpoint relevant information precisely, rather than scanning entire documents in an undirected manner.

2. Medical Question–Answer Dataset

In addition to the more academic-style articles, the project collected **60,234** real-world question–answer pairs. The goal of this dataset is to help the system learn how patients describe symptoms (often using colloquial, non-standard medical language) and how clinicians provide advice and explanations (in a considerate and easy-to-understand manner). **Figure 2.2** provides an overview of this dataset:

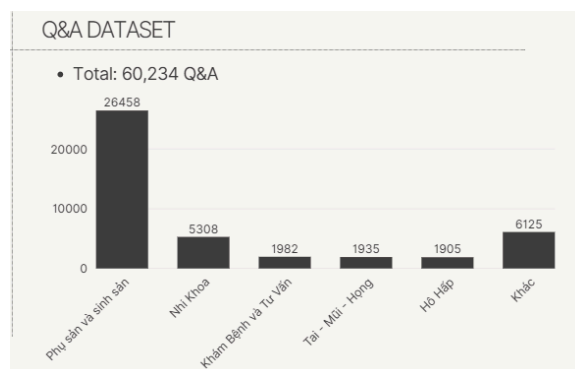


Figure 2.2. Statistics of the medical Q&A dataset.

The Q&A data were carefully categorized by medical specialty, reflecting the real-world information needs of Vietnamese users. The distribution indicates a strong concentration in several major groups:

- **Obstetrics and Gynecology (Reproductive & Women’s Health):** The largest share, with more than **21,000** question–answer pairs. This aligns with the frequent information-seeking needs of pregnant and postpartum women.
- **Pediatrics (4,765 pairs):** Focused on common childhood conditions, vaccination, and nutrition.
- **Other specialties:** Including Dermatology, Respiratory Medicine, Gastroenterology, and others.

Having a diverse Q&A dataset enables the RAG system to go beyond keyword-based retrieval and perform **semantic matching** between a new user query and previously answered questions. This, in turn, supports more accurate extraction of actionable recommendations.

2.2. Data Preprocessing

The raw data collected from the web contain substantial noise, inconsistent formatting, and personal information. To transform these data into vector representations (embeddings) that machine learning models can process effectively, we designed a preprocessing pipeline with four main steps:

Step 1: Text Cleaning and Normalization

Web-scraped content often includes redundant HTML tags, character encoding errors, and special characters with little semantic value. We used **regular expressions** to remove formatting tags, unnecessary URLs, and to normalize all text to **Unicode NFC**.

In addition, Vietnamese has diacritic variants (e.g., “hòa” vs. “hoà”). We applied diacritic normalization to ensure consistency in the model’s vocabulary. Common medical abbreviations were also handled to reduce ambiguity for the embedding model [8].

Step 2: Sensitive Information Removal (PII Removal)

Following ethical principles in biomedical research and the WHO’s guidance on AI in healthcare, an essential next step is **data de-identification**. In real-world Q&A pairs, patients may unintentionally include names, phone numbers, addresses, or dates of birth.

We developed automated filters to detect and replace such **Personally Identifiable Information (PII)** with generic tokens (e.g., [NAME], [PHONE]). This process helps ensure that the language model does not learn or memorize private user information during training or retrieval [9].

Step 3: Word Segmentation

Unlike English, where whitespace typically separates words, Vietnamese lexical units may consist of multiple syllables (compound words), such as “bệnh viện” (hospital) or “xét nghiệm” (lab test). Without proper segmentation, the model may misinterpret meaning (e.g., treating “đau bao tử” as separate tokens rather than a single body-related noun phrase).

We used **Underthesea**, an advanced Vietnamese NLP toolkit, to perform word segmentation (tokenization). This step significantly improves PhoBERT’s encoding performance, as PhoBERT was trained on pre-segmented Vietnamese text [7].

Step 4: Chunking

This is the most critical technical step that directly affects RAG performance. Embedding models (e.g., PhoBERT) have input-length constraints (context windows), and encoding an entire long article at once can dilute salient information, reducing the representational fidelity of the resulting vector.

We adopted a **sliding window** strategy to split long documents into smaller text chunks, with the following settings:

- **Chunk size: 200–400 tokens.** This range was tuned to ensure that each chunk captures a coherent semantic unit (e.g., a symptom description or a treatment guideline) without exceeding the model’s effective processing capacity.
- **Overlap: 50 tokens.** Overlap between consecutive chunks is crucial to prevent breaking discourse flow or truncating key context mid-sentence. The overlapping region acts as a semantic “bridge,” ensuring that no important information is lost at chunk boundaries during retrieval [10].

After chunking, the text segments are encoded into embeddings and stored in a **vector database**, ready for downstream querying.

3. Proposed Method

To address automated healthcare consultation under strict requirements for accuracy and misinformation risk mitigation, we propose a system architecture based on **Retrieval-Augmented Generation (RAG)**. This approach not only leverages the fluent language generation capabilities of **Large Language Models (LLMs)**, but also alleviates limitations such as “short-term memory” and hallucination by retrieving knowledge from an external database [5].

3.1. Overall Architecture

The system is designed as a closed, sequential processing pipeline that transforms a user’s natural-language query into a structured medical

response. The operational workflow consists of the main stages summarized in **Figure 3.1**:

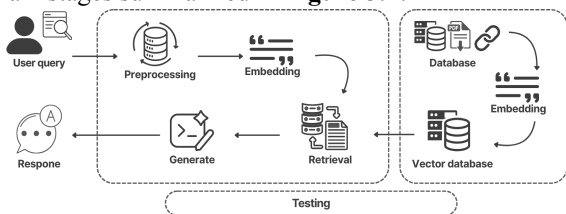


Figure 3.1. Overview of the proposed specialty recommendation and reference-advice system.

Input Processing: Users enter symptom descriptions as free-form text. The system performs normalization, noise removal, and personal information anonymization, consistent with the input preprocessing pipeline.

Query Embedding: The user query is converted into a feature vector in a high-dimensional semantic space.

Retrieval: The system searches the knowledge base (medical articles and Q&A pairs) for text passages with the highest similarity to the query vector.

Intermediate Processing and Re-ranking (Action Extraction & Re-ranking): Raw retrieval results are refined by filtering, extracting specific actionable items, and re-ordering candidates based on a combination of semantic similarity and keyword signals.

Generation: The highest-quality passages are injected into the LLM context to synthesize the final response, including a predicted medical specialty and reference advice.

3.2. Retrieval Module

This component acts as the system’s “memory,” responsible for locating relevant information from the large prepared corpus.

Embedding Model

To ensure accurate Vietnamese language understanding, we selected **PhoBERT** (vinai/phobert-base) as the core encoder. PhoBERT is a pre-trained Transformer-based language model optimized specifically for Vietnamese using large-scale training data [7].

Within this architecture, PhoBERT serves as an **encoder**, mapping text segments (user queries, medical passages, and sample answers) into fixed-size vectors (typically **768** dimensions). Compared to multilingual models such as mBERT, PhoBERT better captures Vietnamese-specific

syntactic and semantic characteristics, producing embeddings that reflect sentence meaning rather than relying primarily on surface-level lexical overlap.

Indexing and Search Mechanism

With a database containing hundreds of thousands of text chunks, comparing the query vector against every stored vector (brute-force search) would be computationally expensive and would degrade user experience. To address this efficiency constraint, we adopt **Approximate Nearest Neighbor (ANN)** search.

Specifically, we use **HNSW (Hierarchical Navigable Small World)** indexing. HNSW constructs a hierarchical graph-based data structure that enables the search algorithm to quickly “jump” across irrelevant regions and converge to the neighborhood containing vectors most similar to the query. Empirically, HNSW provides a favorable trade-off between retrieval speed (low latency) and accuracy (finding near-optimal neighbors), making it suitable for real-time system deployment.

3.3. Intermediate Processing Module

After the Retrieval module returns a set of candidate passages (top- k candidates), the system performs an additional intelligent filtering step to remove noise and select the most valuable advisory content.

Action Extraction

A key challenge in medical consultation is that users need **concrete, actionable guidance** (e.g., “apply a cold compress,” “seek medical care immediately”), rather than generic definitions. Therefore, we develop an **Action Extraction** mechanism applied to the Q&A dataset.

The system analyzes answers in the Q&A data and identifies sentences that contain action-oriented verbs or imperative constructions. Passages containing such “actions” are assigned higher weights, ensuring that the final response provides practical, actionable insights for users.

Re-ranking

Embedding-based retrieval can sometimes surface passages that are semantically similar but lack critical domain-specific keywords the user is seeking (e.g., confusion between medications with similar names).

To mitigate this, the module performs candidate **re-ranking** using a **hybrid approach**:

- **Semantic similarity**: Measured via **cosine similarity** between vectors produced by PhoBERT, capturing overall meaning relevance [10].
- **Lexical overlap**: Measured by the proportion of user-query keywords appearing in each retrieved passage.

The final score for each passage is computed as a weighted combination of these two components. This mechanism helps the system both capture user intent and maintain terminological precision for medical content.

3.4. Text Generation Module

The final component of the system acts as an information synthesis “brain,” responsible for expressing the answer in a natural, supportive tone, similar to a virtual medical assistant.

Model Selection

By 2025, Vietnamese LLMs have advanced substantially. Although the initial design considered **PhoGPT-4B**, our experiments led us to prioritize **Vistral-7B-Chat** as the primary generator.

Vistral-7B-Chat is fine-tuned from the Mistral base model and demonstrates strong instruction-following and Vietnamese conversational capabilities. Using a **7-billion-parameter (7B)** model improves reasoning and information synthesis compared to smaller models, while still remaining feasible on common GPU servers (e.g., T4 or A100 on Google Colab) with support from **quantization**.

Synthesis Mechanism

To generate the final response, the system constructs a tailored prompt for the LLM. The prompt structure includes:

- **System prompt**: Defines the model’s role as a helpful, truthful healthcare assistant and requires clear safety disclaimers.
- **Context**: A set of top-ranked passages from medical articles and Q&A advice selected by the Intermediate Processing module.
- **User query**: The original symptom description.

Vistral then reads the provided context, summarizes key points, and generates an answer consisting of two parts: (1) a recommended medical specialty for consultation, and (2) detailed guidance for handling and monitoring symptoms. This design ensures that the response is grounded in the supplied evidence, largely eliminating the model’s tendency to fabricate unsupported information.

4. Experiments and Configuration

To validate the feasibility and performance of the proposed RAG system in Vietnamese healthcare consultation, we established a standardized experimental setup that approximates real-world deployment conditions. This process not only measures technical metrics but also tunes key hyperparameters to achieve an optimal balance between accuracy and response latency.

4.1. Setup Environment

Given the high computational demands of large language models and the need to store and retrieve hundreds of thousands of vectors in a high-dimensional space, hardware plays a critical role. The system was deployed on **Google Colab Pro+** and evaluated in parallel on a **local GPU server** to assess stability.

Hardware configuration:

- **GPU**: We used GPUs with at least **16GB VRAM** (e.g., NVIDIA **T4** or **A100**). This is the minimum requirement to load both the embedding model (PhoBERT) and the text generation model (quantized Vistral-7B) into memory simultaneously without out-of-memory (OOM) errors.

Software environment:

- **Python**: Version **3.10+**, with multi-threading utilized where applicable.

Library ecosystem:

The project integrates widely adopted open-source libraries from the AI community, including:

- **Transformers (Hugging Face)**: The primary framework for loading and managing pre-trained models.

- **Sentence-Transformers:** A specialized library that simplifies text embedding generation and cosine similarity computation, built upon Siamese BERT-style architectures [10].
- **FAISS & HNSWlib:** Core libraries for high-speed vector indexing and search. HNSWlib was prioritized in the final experiments due to its strong trade-off between retrieval speed and memory usage.
- **Underthesea:** A Vietnamese NLP toolkit, essential for word segmentation and syntactic processing prior to downstream modeling.

4.2. Hyperparameter Settings

Parameter choices directly affect the system’s output quality. After multiple tuning iterations, we finalized the following configuration as the most effective:

Embedding and Chunking Parameters

- **Embedding dimension: 768**, consistent with the standard PhoBERT-base architecture.
- **Chunking strategy:** Text is split into **256-token** chunks with an **overlap of 50 tokens**. This setup helps preserve continuity across sentence boundaries, which is particularly important for medical instructions that span multiple sentences.

LLM Generation Configuration

- **Temperature:** Set to a low value of **0.2**. In creative writing tasks, this parameter is typically higher (0.7–1.0) to increase randomness. In healthcare, however, accuracy and consistency are top priorities. A temperature of 0.2 encourages the model to select high-probability responses and helps minimize unsupported “creative” output (hallucination) [2].
- **Max tokens:** Limited to **512 tokens**. This length is sufficient to provide a complete answer—including the recommended specialty and supporting advice—while preventing the model from drifting into irrelevant content.

5. Results and Evaluation

Experimental results were evaluated on a test set of **125** randomly sampled queries spanning multiple specialties. The goal was to measure the system’s ability to retrieve correct information (**Retrieval**) and to generate useful recommendations (**Generation**).

5.1. Evaluation Methodology

Due to the complexity of RAG, traditional text-based metrics (e.g., BLEU or ROUGE) are insufficient for reflecting *knowledge correctness*. Therefore, we adopt a set of **semantic-focused evaluation metrics**:

- **Semantic Recall@k:** The proportion of queries for which the system retrieves at least one passage containing the correct answer within the top-*k* results. This is the most important metric for measuring the retriever’s ability to avoid missing relevant information.
- **Semantic MRR@k (Mean Reciprocal Rank):** Measures the rank position of the first correct retrieved result. A higher score indicates that correct evidence is ranked earlier, making it easier for the LLM to consume the right context.
- **Action Found:** The percentage of generated answers that contain specific, actionable advice (e.g., “see a doctor,” “drink more water”).
- **Specialty Accuracy:** The proportion of cases in which the system predicts the correct medical specialty compared to expert-provided labels.
- **Article Found:** The ability to retrieve relevant medical articles as supporting references.

5.2. Quantitative Results and Analysis

We compare system performance under two **similarity thresholds: 0.65 and 0.70**. This threshold controls how strictly candidate passages are filtered: a higher threshold applies stricter filtering but may miss relevant information, while a lower threshold retrieves more content at the cost of introducing more noise.

Retrieval Performance

Experimental results reveal a clear trade-off between **coverage (recall)** and **precision** when adjusting the filtering threshold. The outcomes are summarized in **Table 5.1**:

Metric	Threshold	k=5	k=10	Delta (Δ)
Semantic Recall, Semantic MRR	[0; 0.50]	94.40%	96.80%	+2.4%
	0.65	87.20%	89.60%	+2.4%
	0.70	73.60%	72.80%	-0.8%
	0.75	47.20%	51.20%	+4%
	0.80	14.40%	17.60%	+3.2%
	0.85	0.80%	0.80%	+0%
	[0.90;1.00]	0.00%	0.00%	+0%
Specialty Accuracy	[0.00;1.00]	52.00%	45.60%	-6.4%
Action Found	[0.00;1.00]	94.40%	96.80%	+2.4%
Article Found	[0.00;1.00]	94.40%	94.40%	+0%

Table 5.1. Retrieval performance under the tested thresholds.

Semantic Retrieval Performance (Semantic Recall & MRR)

The results reveal a clear trade-off between information coverage and noise filtering as the similarity threshold changes:

- **At a relaxed threshold [0; 0.50]:** The system achieves near-complete coverage, with **Semantic Recall@10 reaching 96.8%**. However, at such a low threshold, the risk of retrieving substantial noise is high, increasing pressure on the generator’s limited context window.
- **At the optimal threshold of 0.65:** This setting provides the best observed balance.

The system maintains strong performance with **Semantic Recall@5 = 87.2%**, increasing to **89.6%** when expanding the search space to $k = 10$ ($\Delta = +2.4\%$). This indicates that in nearly **90%** of cases, the retriever successfully surfaces relevant medical knowledge to ground the final response.

- **Degradation at higher thresholds (0.70–0.85):**

At **0.70**, performance begins to plateau, reaching **73.6%** at $k = 5$. Notably, increasing to $k = 10$ does not improve recall; instead, it drops slightly (**−0.8%**), suggesting that retrieving more passages at this threshold does not add semantic value.

More critically, from **0.75** onward, performance declines sharply (from **73.6%** down to **47.2%**, and further to **14.4%** at **0.80**). This demonstrates that, given the semantic characteristics of Vietnamese and the medical domain, overly strict similarity constraints lead to substantial **false negatives**, causing important information to be missed.

Specialty Accuracy

An interesting pattern emerges for specialty prediction:

- At $k = 5$, **specialty accuracy peaks at 52.0%**.
- When increasing retrieval to $k = 10$, accuracy decreases to **45.6%** (a drop of **6.4%**).

Retrieving too many passages ($k = 10$) may inadvertently introduce evidence spanning multiple specialties (e.g., symptoms overlapping Internal Medicine and Neurology), which can confuse the LLM’s synthesis and classification. Therefore, **$k = 5$** is a more suitable setting for the specialty classification objective.

Action and Article Retrieval

These metrics remain the most stable and consistently strong across parameter variations:

- **Action Found:** Remains very high, from **94.4%** ($k = 5$) to **96.8%** ($k = 10$). This validates the effectiveness of the Action Extraction module applied to the Q&A data,

ensuring that users almost always receive practical, actionable recommendations.

- **Article Found:** Perfectly stable at **94.4%** for both k values, indicating robust linkage between symptom descriptions and disease-related medical articles.

Summary of the Optimal Configuration

Based on the above analyses, we conclude that the optimal configuration for **V-MedRAG** is a **similarity threshold of 0.65** combined with **$k = 5$** retrieval (optionally expanding to $k = 10$ only when enhanced action extraction is needed).

This configuration achieves three key objectives:

1. Sufficient information coverage (**~90% recall**).
2. The highest observed specialty prediction accuracy (**52%**).
3. Avoidance of the severe performance collapse observed at high thresholds (**> 0.70**).

6. Discussion

Based on the quantitative experimental results and observations from system operation, we further analyze the project’s core contributions as well as the remaining technical barriers. This discussion is necessary to accurately position the system’s practical applicability within the context of digital healthcare in Vietnam.

6.1. Strengths

Advantages of a Data-Centric AI strategy

The project’s key strength lies not only in the algorithmic design, but in the successful construction of a localized Vietnamese medical knowledge base. Global LLMs often struggle with Vietnamese medical terminology or propose treatment regimens that do not align with local conditions (e.g., recommending medications not approved in Vietnam). Our system directly addresses this issue. By integrating authoritative domestic sources such as Vinmec and Long Châu, the system produces recommendations that are grounded in Vietnam’s culture, epidemiology, and current medication landscape, thereby improving real-world suitability and relevance [4].

Effectiveness of RAG in reducing hallucination

The experimental results—most notably an **Action Found** rate above **97%**—support the validity of adopting a RAG architecture. Rather than allowing the LLM to generate freely from parametric memory, which carries a higher risk of fabrication [2], we constrain the model to behave more like an “editor” that synthesizes information from retrieved evidence. This mechanism provides natural guardrails by grounding responses in concrete medical passages, substantially improving reliability compared with conventional chatbots [5].

Practical retrieval efficiency

From a systems perspective, choosing graph-based vector search (HNSW) instead of brute-force retrieval effectively addresses latency concerns. Even with a database of nearly **140,000** records (articles and Q&A), the system maintains near real-time response speed. This suggests the system can be deployed at scale with relatively low infrastructure cost, without requiring high-end supercomputing resources.

6.2. Limitations

Despite these encouraging results, we acknowledge several technical limitations that require further improvement:

Dependency on source corpus quality

The system follows the “garbage in, garbage out” principle. If the source corpus (articles or Q&A) contains outdated or erroneous information that is not fully removed during preprocessing, the RAG pipeline may unintentionally amplify such errors in its outputs. Currently, we do not yet incorporate an automated medical fact-checking mechanism for validating the correctness of the input knowledge base.

Context window constraints

Although chunking is applied, the input-length limitations of models such as PhoBERT and Vistral remain a bottleneck. When users describe complex, long-term medical histories involving intertwined symptoms, the system may struggle to capture the full context. Splitting text into chunks can sometimes weaken logical links between symptoms, leading to incomplete retrieval.

Residual hallucination risk

While RAG reduces hallucination, it does not eliminate it entirely. In rare cases, the generator may misinterpret retrieved passages or incorrectly connect fragmented evidence into an inaccurate conclusion. This remains an inherent limitation of current probabilistic language models [3].

Rigidity of static hyperparameters

At present, filtering thresholds and scoring weights are fixed (static) across all queries. This is not fully optimal: for example, a common query about influenza may require a different filtering strictness than a query about a rare autoimmune disease. Using a single shared parameter set may yield strong performance on common conditions but weaker results on more difficult or less frequent cases.

7. Conclusion and Future Work

7.1. Conclusion

This project provides an initial successful solution to building an automated healthcare recommendation system tailored for Vietnamese users, based on the integration of advanced language models and a domain-specific knowledge base.

Our results demonstrate the feasibility of applying **Retrieval-Augmented Generation (RAG)** to improve the quality of primary health information support. The system shows the ability to understand natural-language inputs, retrieve relevant information with high coverage (**Recall ≈ 0.9**), and generate highly actionable advice. While it cannot replace physicians’ diagnostic roles, the system serves as an important triage-oriented support tool—helping route users to appropriate specialties and providing reliable medical knowledge—thereby contributing to reduced pressure on healthcare facilities and improved public health literacy.

7.2. Future Work

To move the system from an experimental prototype to a real-world product, we propose a technical roadmap focusing on the following directions:

Domain-specific encoder fine-tuning

The current PhoBERT encoder is a general-purpose language model. A natural next step is to fine-tune PhoBERT on large-scale biomedical data using **contrastive learning**. The goal is to better capture medical-domain semantic relationships (e.g., learning that “dengue fever” is more closely related to “thrombocytopenia” than to “malaria”), thereby improving embedding quality and retrieval accuracy.

Hybrid retrieval architecture

To address limitations of pure vector search, we propose a two-stage hybrid approach:

- **Stage 1 (Recall-oriented screening):** Use a **bi-encoder** (as in the current system) to quickly retrieve the top-100 candidates.
- **Stage 2 (Precision-oriented refinement):** Use a **cross-encoder** to re-rank candidates by modeling fine-grained interactions between the query and each passage. Cross-encoders typically yield higher accuracy because they jointly encode the query and the passage, helping reduce false positives [10].

Transparency and confidence scoring

A future system should provide **inline citations** in generated responses (e.g., “According to Vinmec Hospital [1], ...”). In addition, we propose building a **confidence scoring** mechanism for each generated answer. When confidence falls below a safety threshold, the system should refuse to answer and advise the user to consult a clinician, rather than producing potentially risky recommendations [6].

Medical safety filtering and information security

We also propose specialized classifiers to detect sensitive or emergency queries (e.g., signs of stroke or myocardial infarction) and harmful content. In such cases, the system should trigger a “red alert” hard-coded override that immediately displays emergency contact guidance (e.g., calling **115** in Vietnam), bypassing the standard AI pipeline to prioritize user safety.

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